To Work or to Shirk: The Shapiro-Stiglitz Theory

Shapiro, C.; Stiglitz, J.E. (1984). "Equilibrium Unemployment as a Worker Discipline Device". The American Economic Review. 74 (3): 433-444

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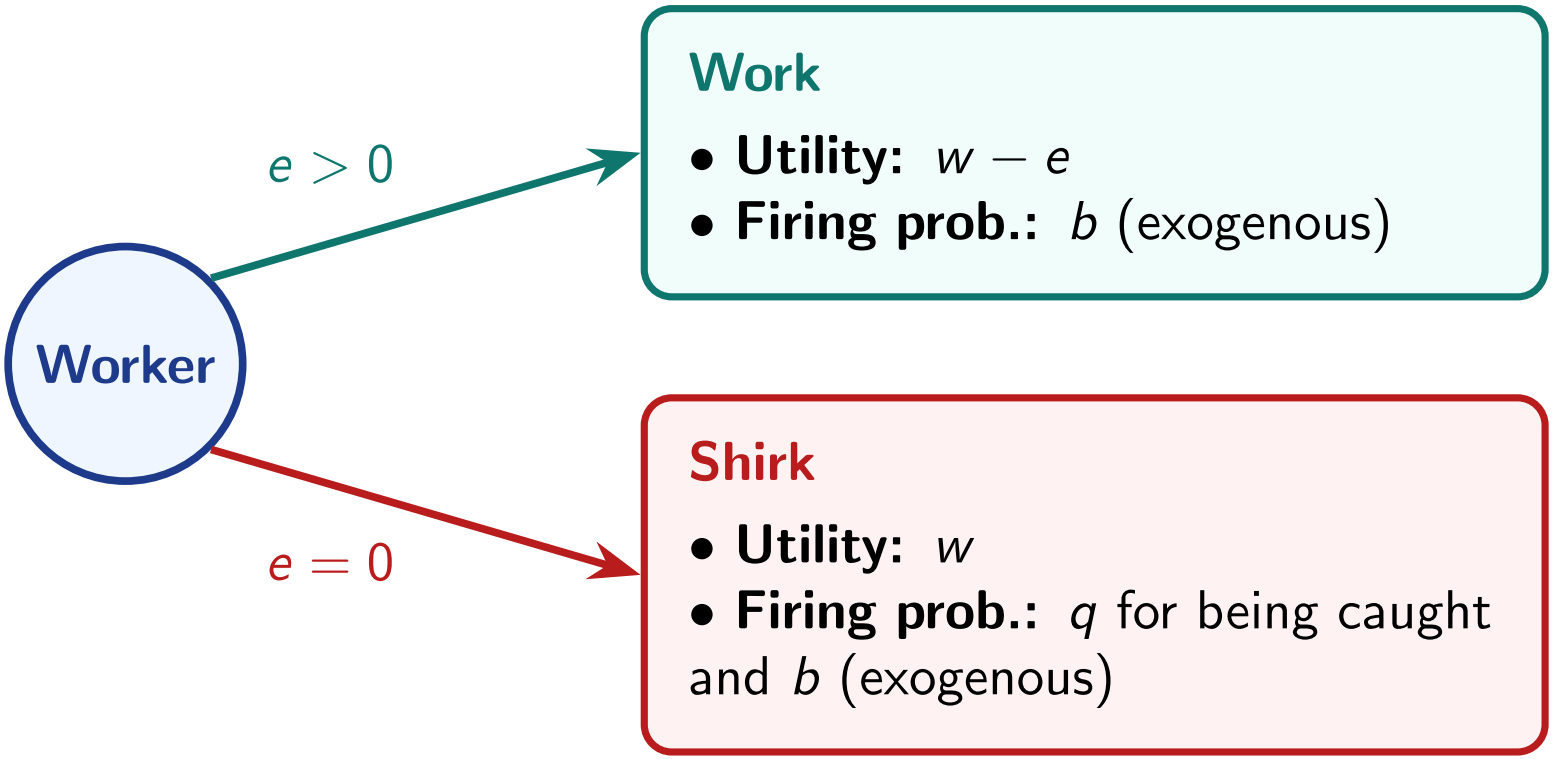
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Worker's Choice

An individual worker's utility function is given by:

$$U(w, e) = w - e$$

where w is the wage and e is the effort exerted by the worker (identical for all workers). The worker has two choices:



Asset Equations I

If she decides to work, her asset equation is given by:

$$\underbrace{rV_E^N}_{\text{Return on Asset}} = \underbrace{w - e}_{\text{Flow Benefit}} + \underbrace{b(V_u - V_E^N)}_{\text{Expected Capital Loss}}$$

- Probability of being fired: b (exogenous)
- Loss: $V_u - V_E^N$

If she decides to shirk, her asset equation is given by:

$$\underbrace{rV_E^S}_{\text{Return on Asset}} = \underbrace{w}_{\text{Flow Benefit}} + \underbrace{(b+q)(V_u - V_E^S)}_{\text{Expected Capital Loss}}$$

- Probability of being fired: $b + q$ (exogenous + endogenous)
- Loss: $V_u - V_E^S$

Asset Equations II

Finally, if she is unemployed, her asset equation is given by:

$$\underbrace{rV_u}_{\text{Return on Asset}} = \underbrace{\bar{w}}_{\text{Flow Benefit}} + \underbrace{a(V_E - V_u)}_{\text{Expected Capital Gain}}$$

- Unemployment benefits: $\bar{w}$
- Hiring rate: a
- Gain: $V_E - V_u$ (where V_E is either V_E^N or V_E^S depending on whether the worker will work or shirk after being hired)

No-Shirking Condition

In order for a worker to exert effort (not shirk), the following condition must hold:

$$V_E^N \geq V_E^S$$

Plugging in the asset equations, we get the **No-Shirking Condition (NSC)**:

$$w \geq rV_u + \frac{(r+b+q)e}{q} \equiv \hat{w}$$

whereby $\hat{w}$ is the *critical wage*.

Firm's Choice

- If $w < \hat{w}$, workers will shirk due to (1) the low wage w and (2) a high probability a of being rehired (high labor demand at low w). $\Rightarrow$ Firm should increase w to $\hat{w}$ to prevent shirking.
- If $w > \hat{w}$, workers will exert effort e due to (1) the high wage w and (2) a low probability a of being rehired (low labor demand at high w). $\Rightarrow$ Firm should decrease w to $\hat{w}$ to save costs.

Conclusion

The *NSC* is binding for the firm, i.e. all firms will choose the lowest possible wage that fulfills it, which is $\hat{w}$. They will also pay the minimum unemployment benefits $\bar{w}$ because higher benefits both directly increase costs and increase the *NSC*.

Aggregate NSC I

Plugging in the asset equation for unemployed workers rV_u, we get the **Aggregate NSC**:

$$w \geq \bar{w} + e + \frac{e(a+b+r)}{q}$$

In steady state, the employment pool L stays constant, i.e. the flow into unemployment must equal the flow out of unemployment:

$$bL = a(N-L) \Leftrightarrow a = \frac{bL}{N-L}$$

where L is the number of employed workers and N is the total number of workers. Now plugging steady state a and V_u into the NSC, we get the **Aggregate Steady-State NSC**:

$$w \geq e + \bar{w} + \frac{e}{q}\left(\frac{b}{u} + r\right) \equiv \hat{w} \quad \text{where } u = \frac{N-L}{N}$$

Aggregate NSC II

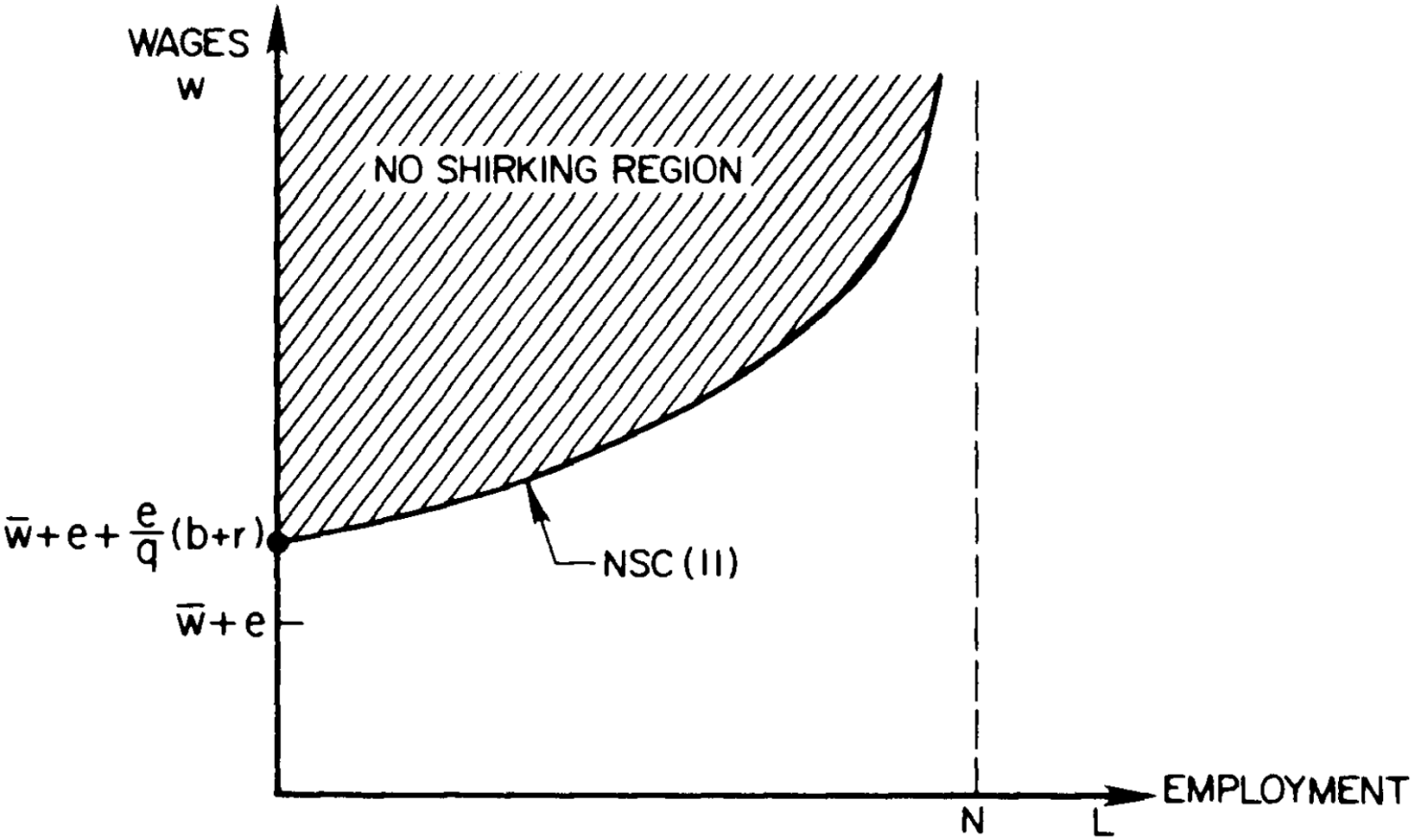


FIGURE 1. THE AGGREGATE NO-SHIRKING CONSTRAINT

Equilibrium

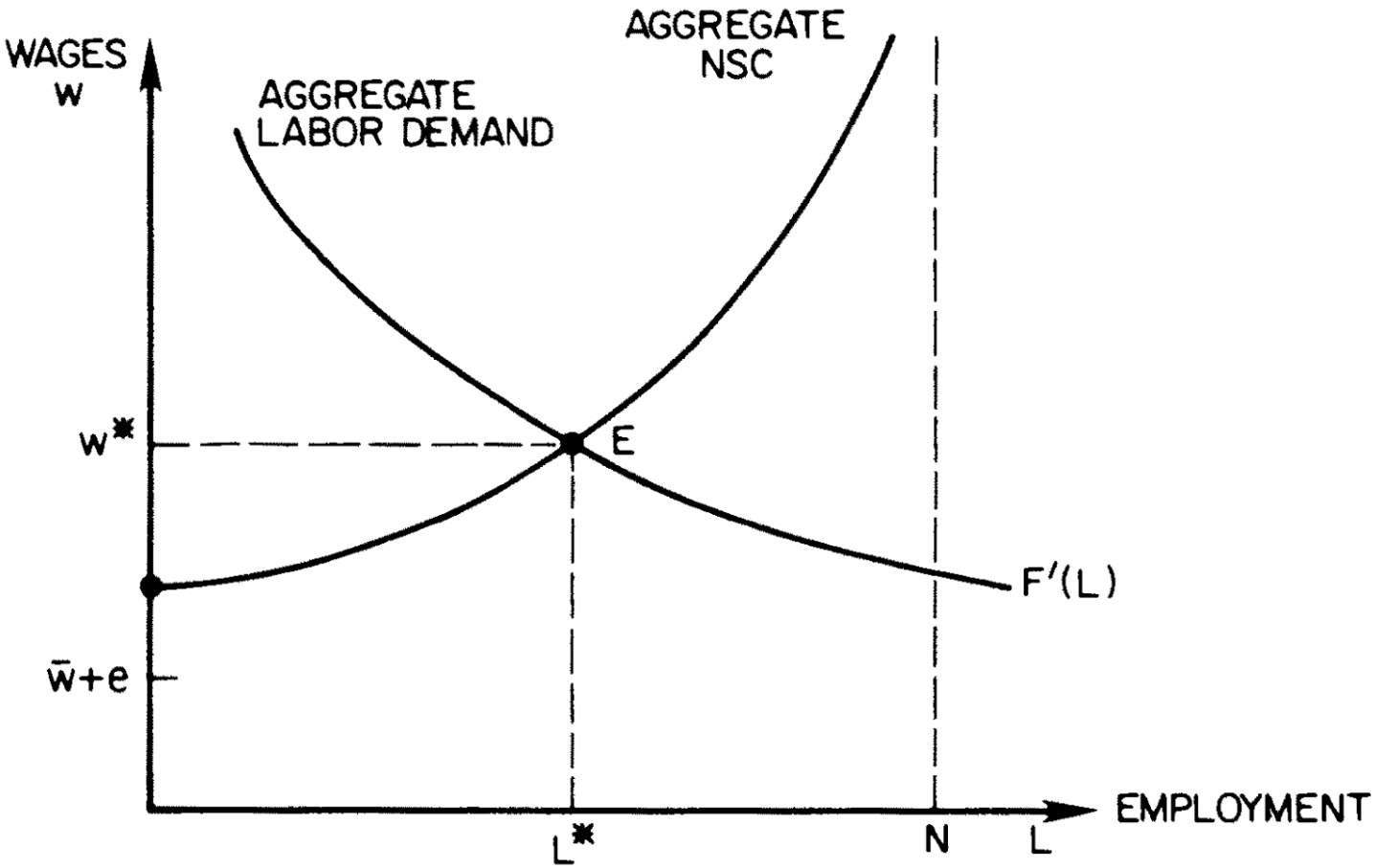


FIGURE 2. EQUILIBRIUM UNEMPLOYMENT

Comparative Statics

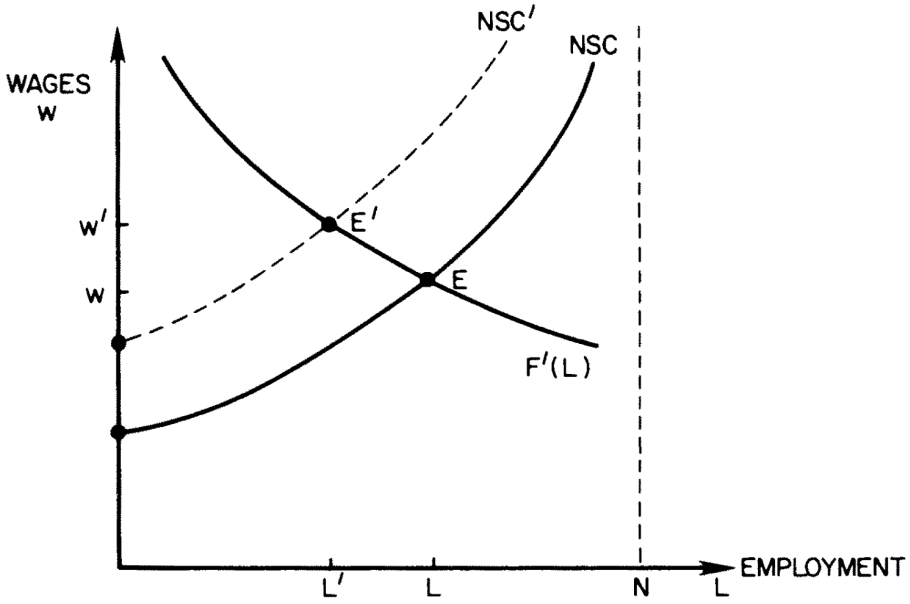


FIGURE 3. COMPARATIVE STATICS

NSC shifts up if:

- quit rate b increases
- monitoring intensity q decreases
- required effort e increases

Welfare Analysis

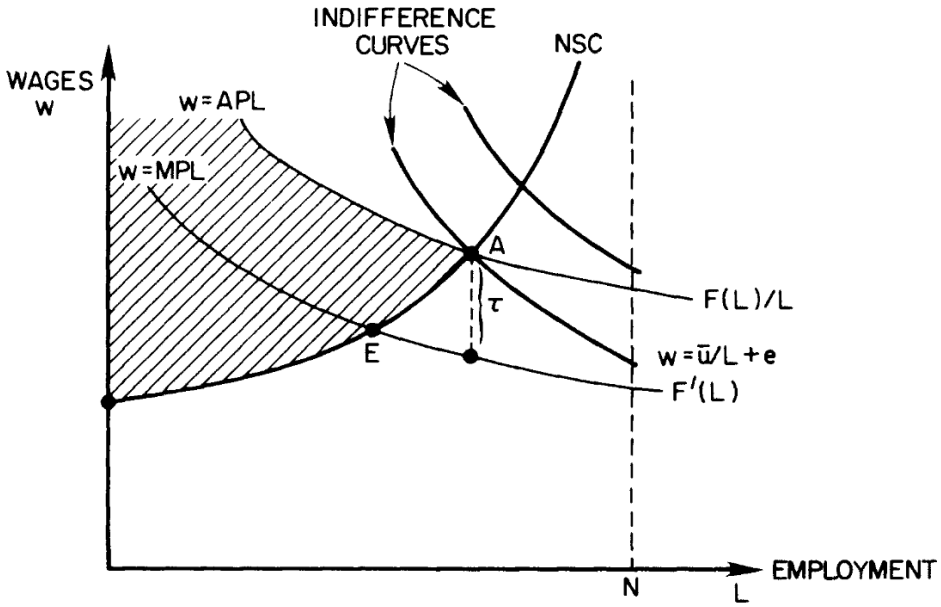


FIGURE 4. SOCIAL OPTIMUM AT *A*

A *wage subsidy* τ (financed by taxing away firm profits) could increase aggregate welfare, employment, and production.

It constitutes a *Pareto improvement* if firms are owned by workers, but not if capital owners and workers are distinct classes.

$\Rightarrow$ **No separation between efficiency and income distribution.**

Discussion

What should be the minimum requirement for unemployment benefits $\bar{w}$? How does it differ based on the risk profile of workers?

What alternative discipline devices could be used to reduce shirking?

Appendix

Sluggish Wage Adjustment After a Demand Shock

